

Lecture 6

Multiplexing

Multiplexing

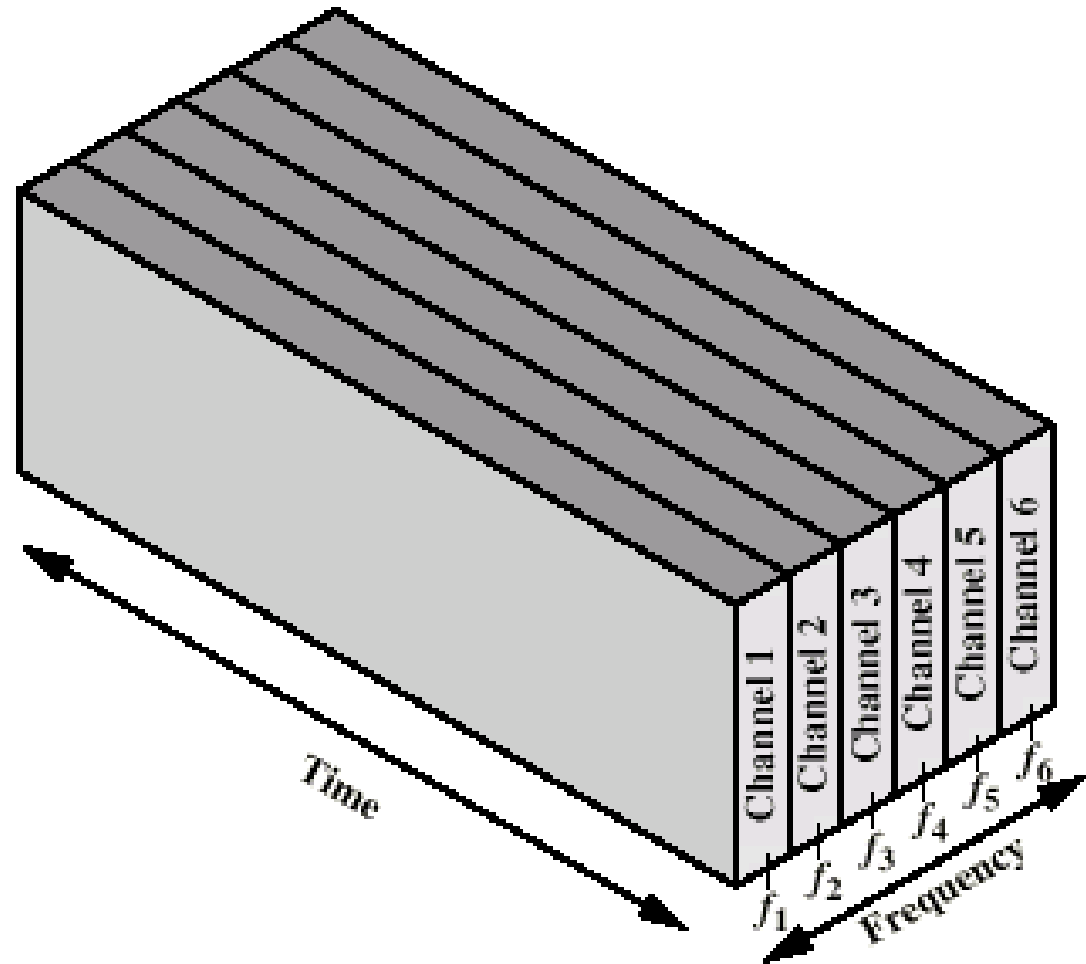
- It set of techniques that allows the simultaneously transmission of multiple signals across a single data link.



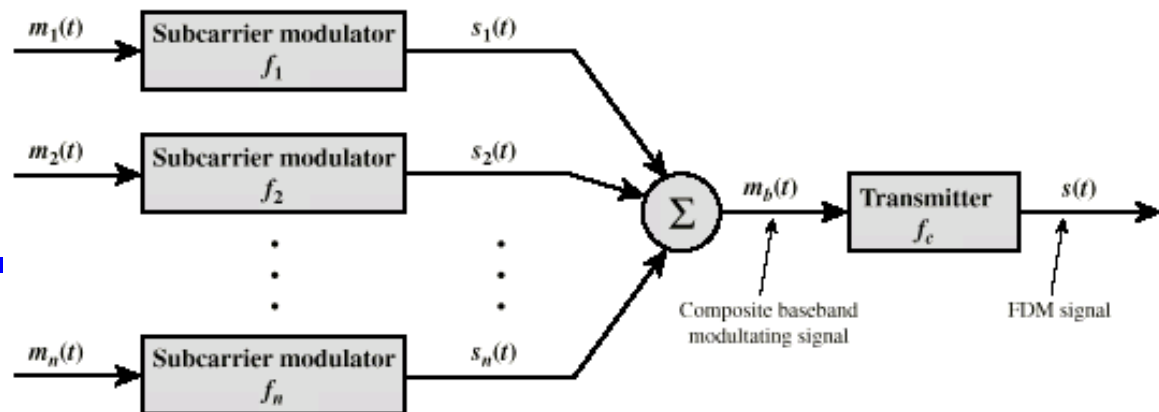
Frequency Division Multiplexing

- FDM
- Useful bandwidth of medium exceeds required bandwidth of channel
- Each signal is modulated to a different carrier frequency
- Carrier frequencies separated so signals do not overlap (guard bands)
- e.g. broadcast radio
- Channel allocated even if no data

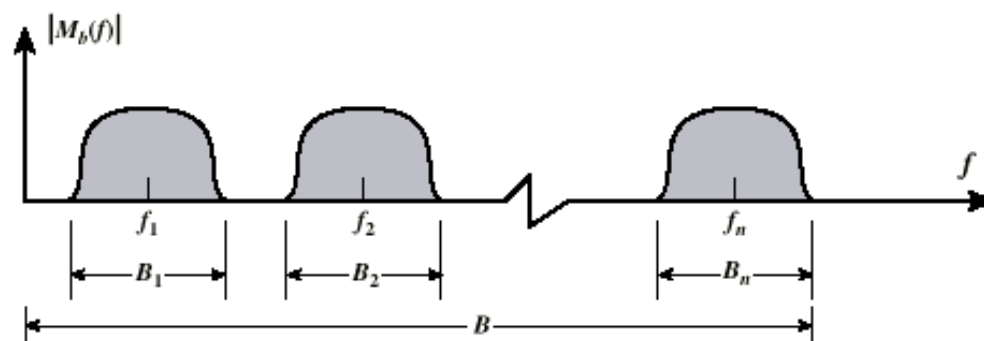
Frequency Division Multiplexing Diagram



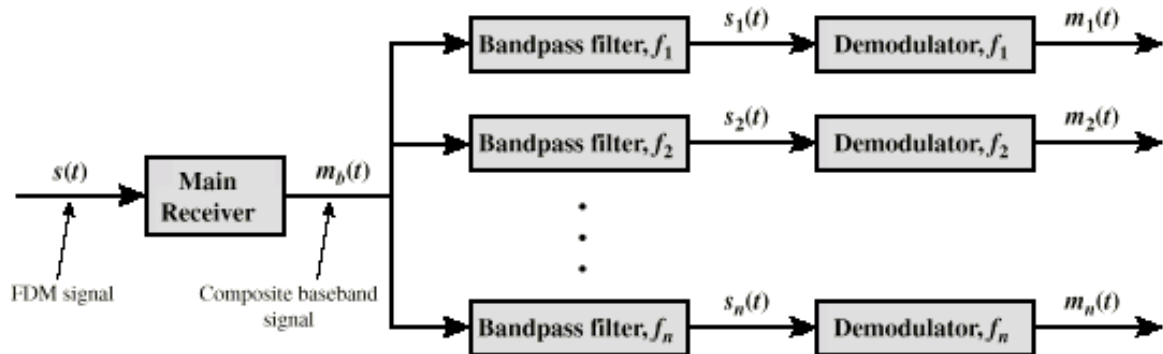
FDM System



(a) Transmitter



(b) Spectrum of composite baseband modulating signal

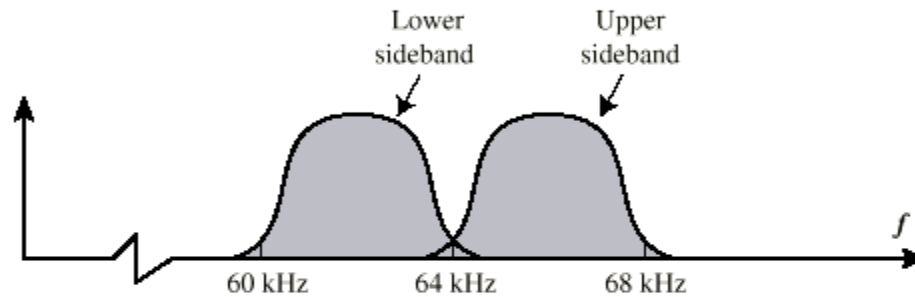


(c) Receiver

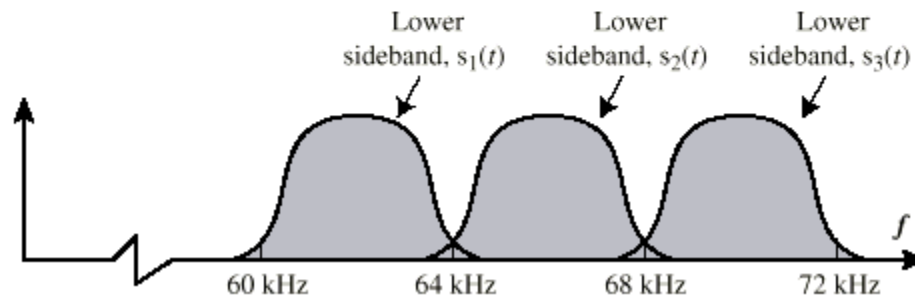
FDM of Three Voiceband Signals



(a) Spectrum of $m_1(t)$, positive f



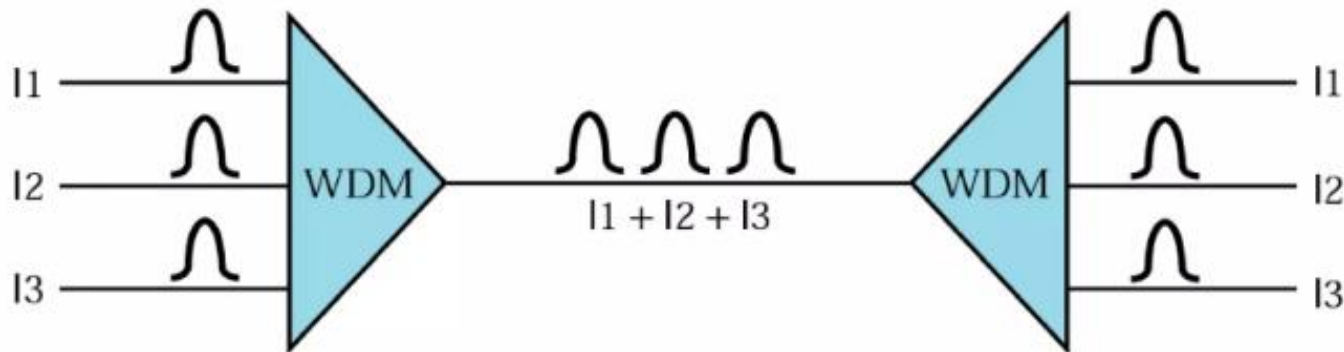
(b) Spectrum of $s_1(t)$ for $f_1 = 64$ kHz



(c) Spectrum of composite signal using subcarriers at 64 kHz, 68 kHz, and 72 kHz

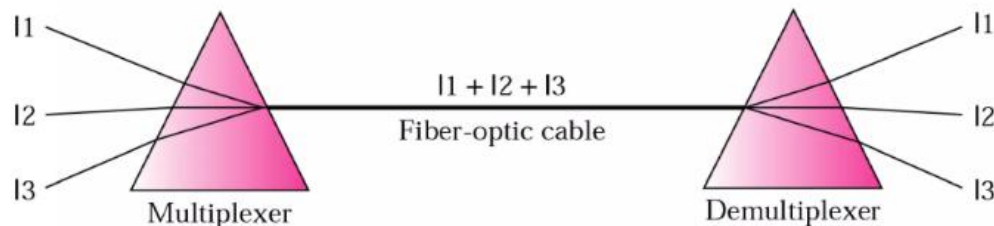
Wavelength Division Multiplexing

- Multiple beams of light at different frequency
- Carried by optical fiber
- A form of FDM
- Each color of light (wavelength) carries separate data channel

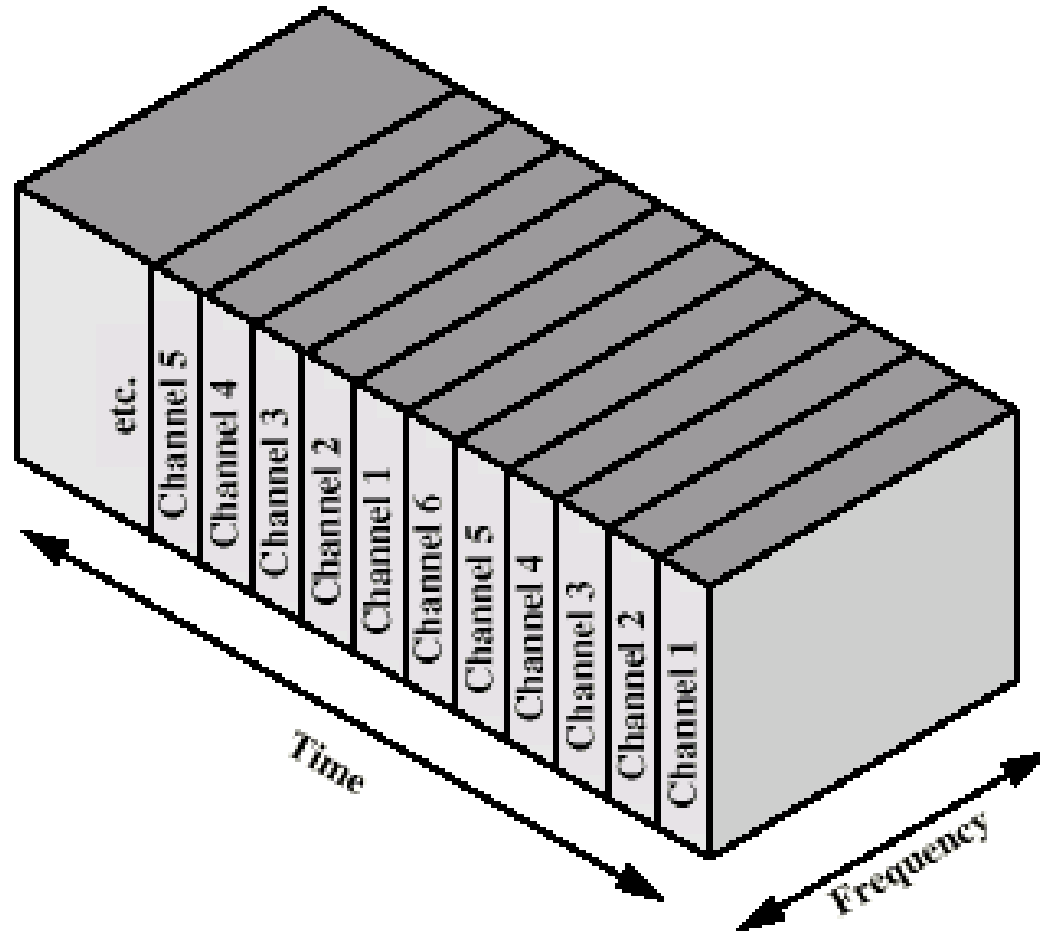


WDM Operation

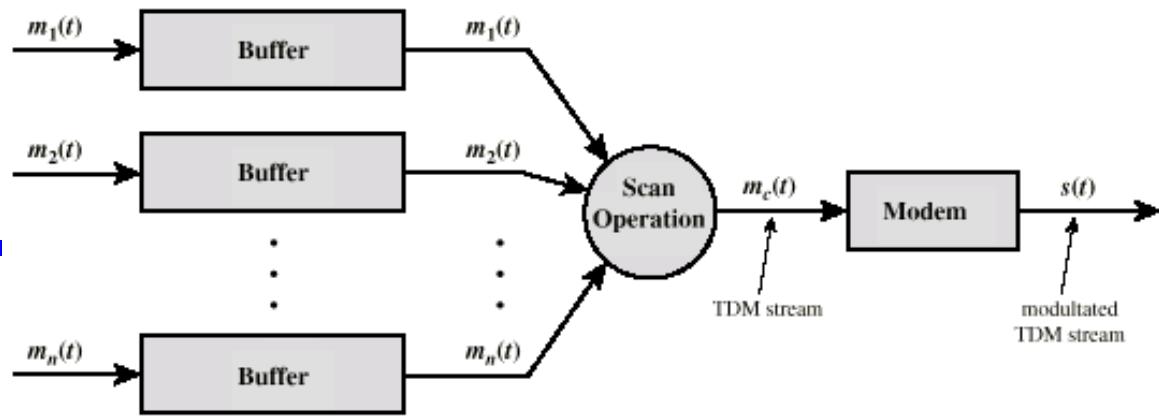
- Same general architecture as other FDM
- Number of sources generating laser beams at different frequencies
- Multiplexer consolidates sources for transmission over single fiber
- Optical amplifiers amplify all wavelengths
 - Typically tens of km apart
- Demux separates channels at the destination



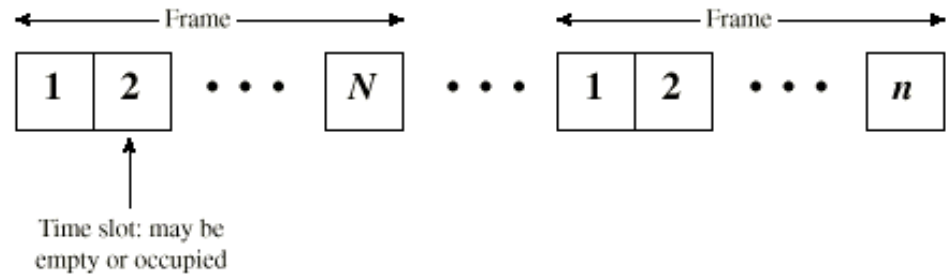
Time Division Multiplexing



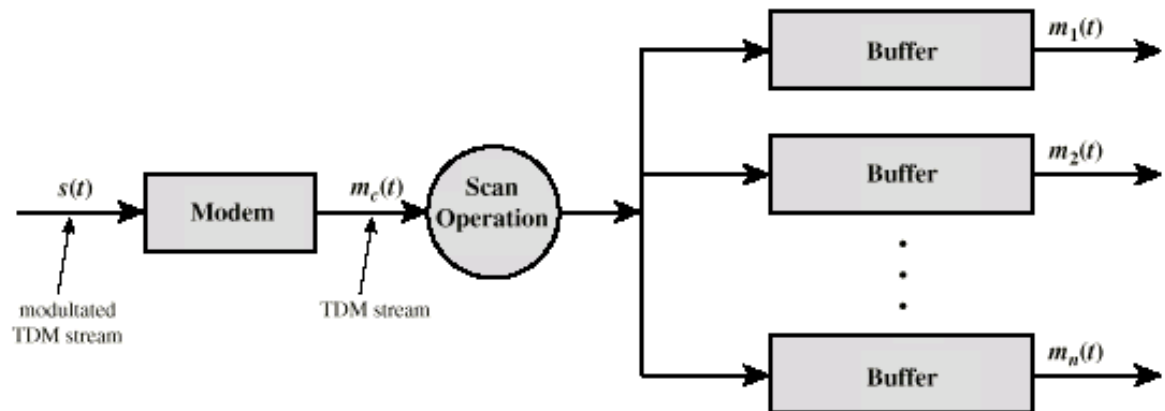
TDM System



(a) Transmitter



(b) TDM Frames



(c) Receiver

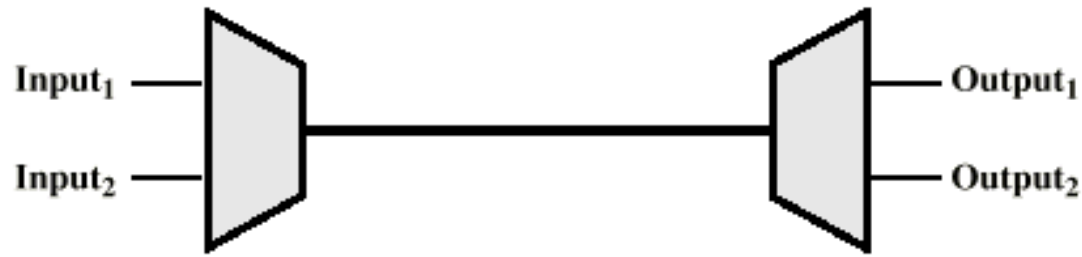
Synchronous Time Division Multiplexing

- Data rate of medium exceeds data rate of digital signal to be transmitted
- Multiple digital signals interleaved in time
- May be at bit level or blocks
- Time slots preassigned to sources and fixed
- Time slots allocated even if no data
- Time slots do not have to be evenly distributed amongst sources

TDM Link Control

- No headers and trailers
- Data link control protocols not needed
- Flow control
 - Data rate of multiplexed line is fixed
 - If one channel receiver can not receive data, the others must carry on
 - The corresponding source must be quenched
 - This leaves empty slots
- Error control
 - Errors are detected and handled by individual channel systems

Data Link Control on TDM



(a) Configuration

Input₁..... F₁ f₁ f₁ d₁ d₁ d₁ C₁ A₁ F₁ f₁ f₁ d₁ d₁ d₁ C₁ A₁ F₁

Input₂... F₂ f₂ f₂ d₂ d₂ d₂ d₂ C₂ A₂ F₂ f₂ f₂ d₂ d₂ d₂ d₂ C₂ A₂ F₂

(b) Input data streams

... f₂ F₁ d₂ f₁ d₂ f₁ d₂ d₁ d₂ d₁ C₂ d₁ A₂ C₁ F₂ A₁ f₂ F₁ f₂ f₁ d₂ f₁ d₂ d₁ d₂ d₁ d₂ d₁ C₂ C₁ A₂ A₁ F₂ F₁

(c) Multiplexed data stream

Legend: F = flag field d = one octet of data field
 A = address field f = one octet of FCS field
 C = control field

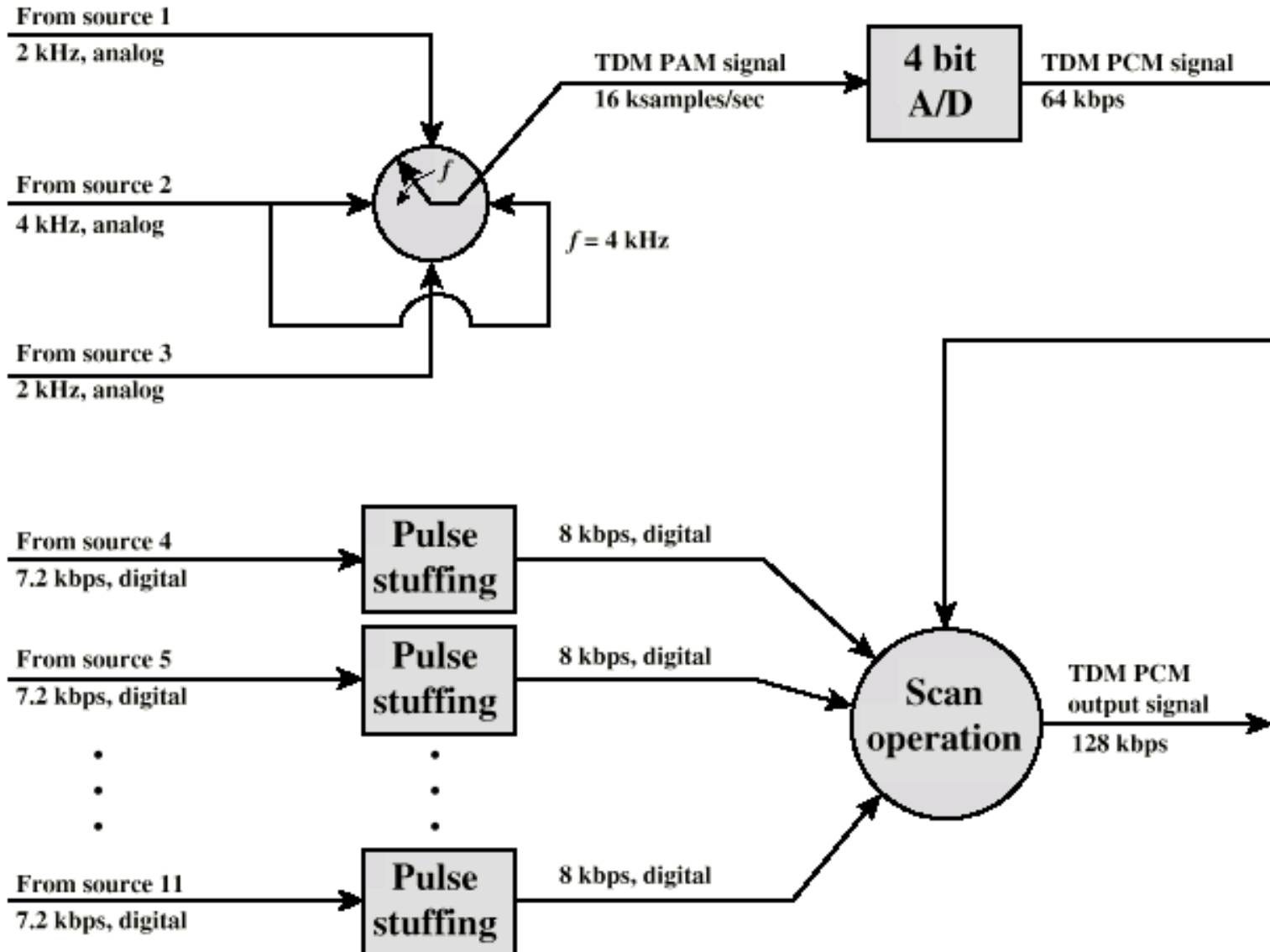
Framing

- No flag or SYNC characters bracketing TDM frames
- Must provide synchronizing mechanism
- Added digit framing
 - One control bit added to each TDM frame
 - Looks like another channel - “control channel”
 - Identifiable bit pattern used on control channel
 - e.g. alternating 01010101...unlikely on a data channel
 - Can compare incoming bit patterns on each channel with sync pattern

Pulse Stuffing

- Problem - Synchronizing data sources
- Clocks in different sources drifting
- Data rates from different sources not related by simple rational number
- Solution - Pulse Stuffing
 - Outgoing data rate (excluding framing bits) higher than sum of incoming rates
 - Stuff extra dummy bits or pulses into each incoming signal until it matches local clock
 - Stuffed pulses inserted at fixed locations in frame and removed at demultiplexer

TDM of Analog and Digital Sources



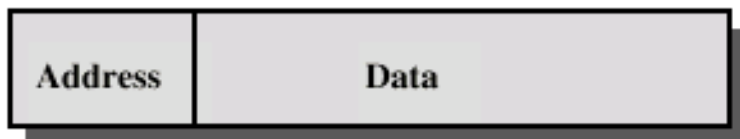
Statistical TDM

- In Synchronous TDM many slots are wasted
- Statistical TDM allocates time slots dynamically based on demand
- Multiplexer scans input lines and collects data until frame full
- Data rate on line lower than aggregate rates of input lines

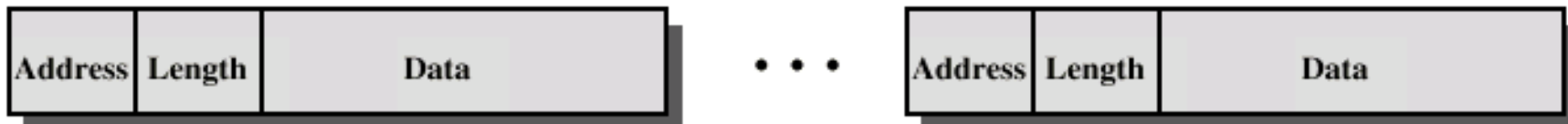
Statistical TDM Frame Formats



(a) Overall frame



(b) Subframe with one source per frame



(c) Subframe with multiple sources per frame

Performance

- Output data rate less than aggregate input rates
- May cause problems during peak periods
 - Buffer inputs
 - Keep buffer size to minimum to reduce delay

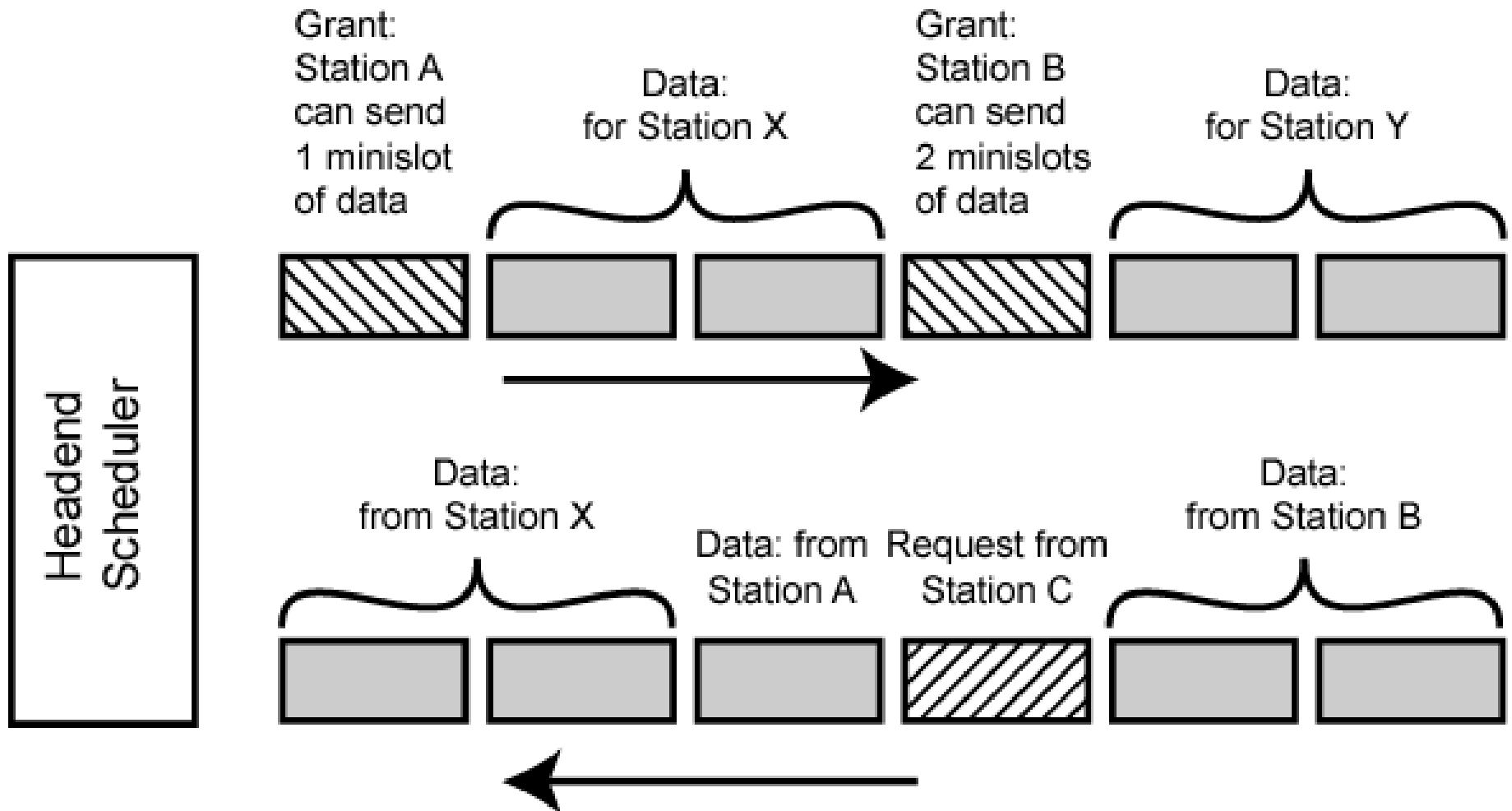
Cable Modem Outline

- Two channels from cable TV provider dedicated to data transfer
 - One in each direction
- Each channel shared by number of subscribers
 - Scheme needed to allocate capacity
 - Statistical TDM

Cable Modem Operation

- Downstream
 - Cable scheduler delivers data in small packets
 - If more than one subscriber active, each gets fraction of downstream capacity
 - May get 500kbps to 1.5Mbps
 - Also used to allocate upstream time slots to subscribers
- Upstream
 - User requests timeslots on shared upstream channel
 - Dedicated slots for this
 - Headend scheduler sends back assignment of future time slots to subscriber

Cable Modem Scheme



Thank You